

Lattice Energy

Question Paper

Level	A Level
Subject	Chemistry
Exam Board	Edexcel
Topic	Energetics II
Sub Topic	Lattice Energy
Booklet	Question Paper
Paper Type	Multiple Choice

Time Allowed: 26 minutes

Score: /21

Percentage: /100

Grade Boundaries:

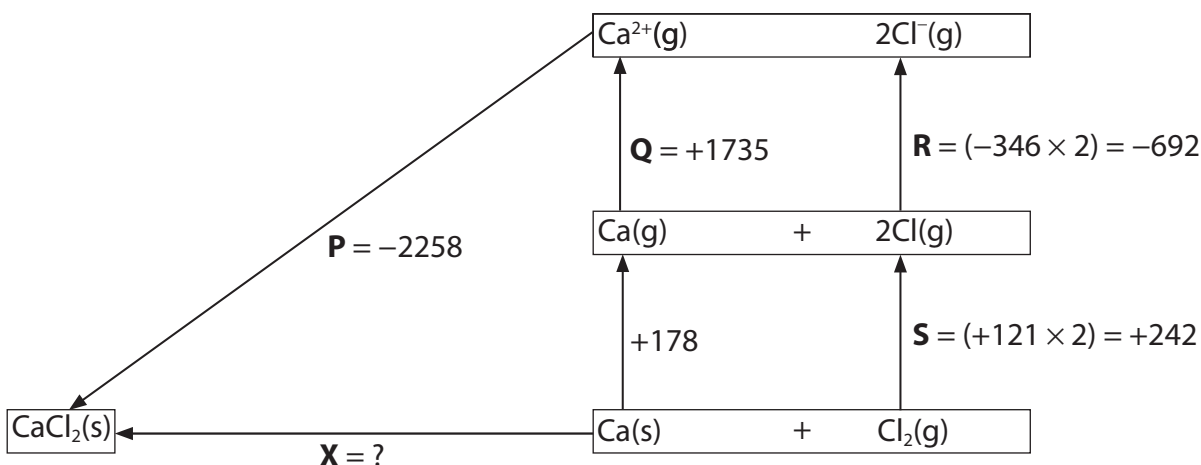
A*	A	B	C	D	E	U
>85%	77.5%	70%	62.5%	57.5%	45%	<45%

1 What is the equation for the first electron affinity of sulfur?

- ☐ A $\text{S(s)} + \text{e}^- \rightarrow \text{S}^-(\text{g})$
- ☐ B $\text{S(g)} + \text{e}^- \rightarrow \text{S}^-(\text{g})$
- ☐ C $\text{S(s)} \rightarrow \text{S}^+(\text{g}) + \text{e}^-$
- ☐ D $\text{S(g)} \rightarrow \text{S}^+(\text{g}) + \text{e}^-$

(Total for Question = 1 mark)

2 The diagram shows a Born-Haber cycle for calcium chloride. It is not drawn to scale. All units are in kJ mol^{-1} .



(a) Which enthalpy change is correctly labelled on the diagram?

(1)

- ☐ A Enthalpy change for the formation of calcium chloride (**P**).
- ☐ B First ionization energy of calcium (**Q**).
- ☐ C Electron affinity of chlorine (**R**).
- ☐ D Twice the enthalpy change of atomization of chlorine (**S**).

(b) What is the value of **X**, in kJ mol^{-1} ?

(1)

- ☐ A +795
- ☐ B -795
- ☐ C +3721
- ☐ D -3721

(Total for Question = 2 marks)

3 Which of the following data is **not** needed to calculate the lattice energy of sodium chloride when using a Born-Haber cycle?

- ☐ A Enthalpy change of formation of sodium chloride.
- ☐ B Enthalpy change of atomization of sodium.
- ☐ C First ionization energy of chlorine.
- ☐ D Electron affinity of chlorine.

(Total for Question = 1 mark)

4 The lattice energy of magnesium oxide is more negative than the lattice energy of magnesium fluoride because

- ☐ A oxide ions are larger than fluoride ions.
- ☐ B oxide ions are larger than magnesium ions.
- ☐ C oxide ions are more highly charged than fluoride ions.
- ☐ D there is only one oxide ion but two fluoride ions per magnesium ion.

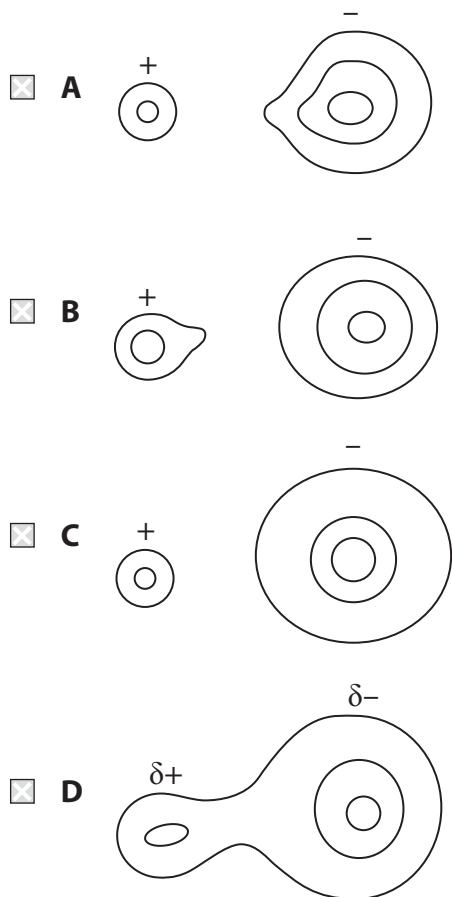
(Total for Question = 1 mark)

- 5 Which of the following quantities, used in the calculation of the lattice energy of lithium oxide, Li_2O , has a negative value?

- ☐ A The enthalpy change of atomization of lithium.
- ☐ B The first ionization energy of lithium.
- ☐ C The first electron affinity of oxygen.
- ☐ D The second electron affinity of oxygen.

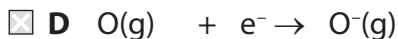
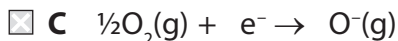
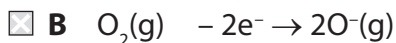
(Total for Question = 1 mark)

- 6 Which of the diagrams below best represents the shapes of the electron contours in sodium fluoride?



(Total for Question = 1 mark)

7 Which of the equations below represents the first electron affinity for oxygen?



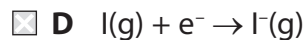
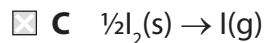
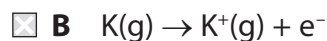
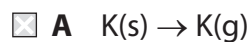
(Total for Question = 1 mark)

8 Which of the following oxides would be expected to have the most exothermic lattice energy?



(Total for Question = 1 mark)

9 In the Born-Haber cycle for potassium iodide, which of the following steps is **exothermic**?



(Total for Question = 1 mark)

- 10 Magnesium chloride, MgCl_2 , has two lattice energy values quoted in the data booklet. The first is the experimental value, obtained from the Born-Haber cycle, $-2526 \text{ kJ mol}^{-1}$; the second is the theoretical value, $-2326 \text{ kJ mol}^{-1}$. Why are the two values different?

- ☐ A The cation polarizes the anion leading to some covalent bonding.
- ☐ B The anion polarizes the cation leading to some covalent bonding.
- ☐ C Magnesium chloride is a covalent substance.
- ☐ D The results from the Born-Haber cycle are too inaccurate to be reliable.

(Total for Question = 1 mark)

- 11 Which of the following represents the process occurring when the enthalpy change of atomization of bromine is measured?

- ☐ A $\frac{1}{2}\text{Br}_2(\text{l}) \rightarrow \text{Br}(\text{g})$
- ☐ B $\frac{1}{2}\text{Br}_2(\text{g}) \rightarrow \text{Br}(\text{g})$
- ☐ C $\text{Br}_2(\text{l}) \rightarrow \text{Br}^+(\text{g}) + \text{Br}^-(\text{g})$
- ☐ D $\text{Br}_2(\text{g}) \rightarrow \text{Br}^+(\text{g}) + \text{Br}^-(\text{g})$

(Total for Question = 1 mark)

- 12 The standard enthalpy changes of formation of some sulfur species are:

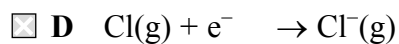
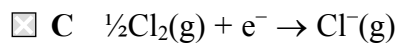
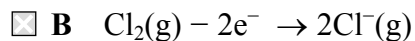
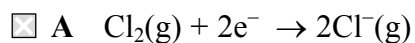
Species	$\Delta H_f^\ominus / \text{kJ mol}^{-1}$
$\text{S}_8(\text{s})$	0
$\text{S}_8(\text{g})$	+103
$\text{S}(\text{g})$	+279

The enthalpy of atomization of sulfur is (in kJ mol^{-1})

- ☐ A 103
- ☐ B 279
- ☐ C 279
- ☐ D $(103 \div 8) + 279$

(Total for Question = 1 mark)

13 Which of these equations represents the electron affinity of chlorine?



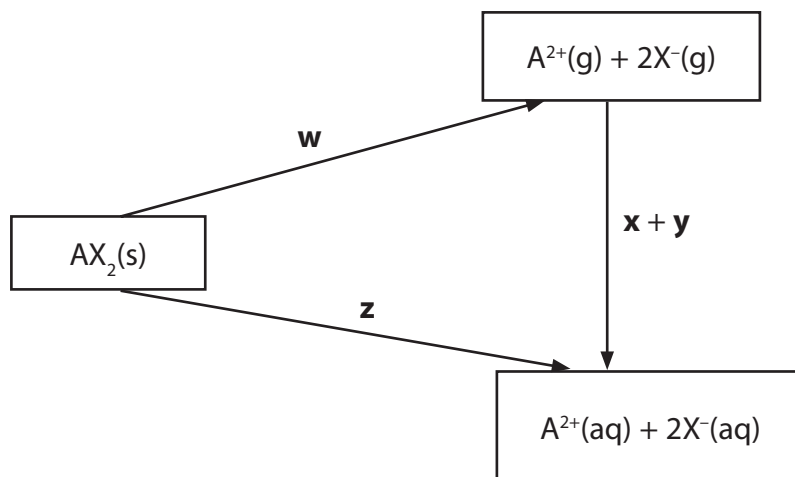
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14 Which of these ions has the greatest ability to polarize an anion?



(Total for Question = 1 mark)

- 15 The following cycle represents the enthalpy changes **w**, **x**, **y** and **z**, occurring when an ionic solute, $AX_2(s)$, dissolves in water.



Which of the changes is the lattice energy of $AX_2(s)$?

- ☐ A $\frac{1}{2} w$
- ☐ B $-w$
- ☐ C z
- ☐ D $z - x - y$

(Total for Question = 1 mark)

- 16 The enthalpy change of atomization of iodine is the value of ΔH for the process

- ☐ A $I_2(s) \rightarrow I_2(g)$
- ☐ B $I_2(s) \rightarrow 2I(g)$
- ☐ C $I_2(g) \rightarrow 2I(g)$
- ☐ D $\frac{1}{2}I_2(s) \rightarrow I(g)$

(Total for Question = 1 mark)

17 The ionic radii in nm of some ions are given below.

Li^+	0.074	F^-	0.133
Ca^{2+}	0.100	Cl^-	0.180
		O^{2-}	0.140
		S^{2-}	0.185

(a) Which of the following compounds has the most exothermic lattice energy? They all have the same crystal structure.

(1)

- ☐ A LiF
- ☐ B LiCl
- ☐ C CaO
- ☐ D CaS

(b) Which of the following compounds will show the greatest difference between the experimental (Born-Haber) lattice energy and that calculated from a purely ionic model?

(1)

- ☐ A LiF
- ☐ B Li_2O
- ☐ C CaO
- ☐ D CaS

(Total for Question = 2 marks)

18 The equation for the enthalpy of hydration for a magnesium ion is

- ☐ **A** $\text{Mg}^{2+}(\text{s}) + \text{aq} \rightarrow \text{Mg}^{2+}(\text{aq})$
- ☐ **B** $\text{Mg}^{2+}(\text{g}) + \text{aq} \rightarrow \text{Mg}^{2+}(\text{aq})$
- ☐ **C** $\text{Mg}^{2+}(\text{aq}) \rightarrow \text{Mg}^{2+}(\text{g}) + \text{aq}$
- ☐ **D** $\text{Mg}^{2+}(\text{aq}) \rightarrow \text{Mg}^{2+}(\text{s}) + \text{aq}$

(Total for Question = 1 mark)

19 Which reaction has an enthalpy change equal to the enthalpy of hydration of the sodium ion?

- ☐ **A** $\text{Na}^+(\text{g}) + \text{excess H}_2\text{O}(\text{l}) \rightarrow \text{Na}^+(\text{aq})$
- ☐ **B** $\text{Na}^+(\text{g}) + 1 \text{ mol of H}_2\text{O}(\text{l}) \rightarrow \text{Na}^+(\text{aq})$
- ☐ **C** $\text{Na}^+(\text{s}) + \text{excess H}_2\text{O}(\text{l}) \rightarrow \text{Na}^+(\text{aq})$
- ☐ **D** $\text{Na}^+(\text{s}) + 1 \text{ mol of H}_2\text{O}(\text{l}) \rightarrow \text{Na}^+(\text{aq})$

(Total for Question = 1 mark)